

The Effect of Alternative Scaling Approaches on the Performance of Different Supervised Learning Algorithms. An Empirical Study in the Case of Credit Scoring

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Abstract

Building classification tools to discriminate between good and bad credit risks is a supervised learning task that can be solved using different approaches. In constructing such tools, generally, a set of training data containing qualitative and quantitative attributes is used to learn the discriminant rules. In real world of credit applications a lot of the available information about the customer and his payment behavior appears in qualitative categorical attributes.

On the other hand many approaches of supervised learning require quantitative numerical input attributes. Qualitative attributes first have to be transformed in numerical, before they can be used for the learning process.

One very simple approach to handle that problem is to code each possible value of all qualitative categorical attributes in new separate binary attributes. This leads to an increasing number of input attributes, that makes learning more complicated and less reliable. In particular neural networks need more time for training and often loose accuracy.

In this paper we consider different scaling approaches - here the number of attributes does not increase - to transform categorical into numerical attributes. We use them as input variables to learn the discriminant rules in order to enhance accuracy and stability of the rules. Using real world credit data, we evaluate different approaches and compare the results.

Introduction

Looking at the situation in credit business over the last years we realize that things fundamentally changed: On the one hand a lot of companies (e.g. leasing companies, mail-order firms, service providers in cellular phone business, etc.) joined the credit market not belonging to the conventional circle of lenders (banks in general). All these companies rely on complete information about the credibility of their customers to be protected against loss in case of bad credit risks. On the other hand each credit lender plays on the market among other competitors, so there is a strong interest to decide on accepting or rejecting of (unknown) customers as immediate as possible. Credit scoring aims to develop an objective methodology leading to reproducible results to support credit officers.

In this paper we evaluate the effect of different scaling approaches on the performance of various supervised

learning algorithms regarding the fact that a lot of these algorithms can not handle categorical attributes, directly. In credit business, however, a lot of the available information about customers and their behavior of payment is categorical, so data transformation is necessary. In a first step we consider credit scoring as a classification problem and describe in summary different supervised learning algorithms that can deal with this problem. Thereafter we discuss the alternative transformation methods, among them coding and scaling approaches. In a further step we present and analyze our experimental results on a real credit domain. The last section is devoted to our conclusions.

Credit Scoring - A Classification Problem

The task of credit scoring can be described, formally, as a classification problem: Using the information the customer gives by filling out the credit application form (input variables) we have to discriminate him in certain predefined risk class (unknown output variable). On the other hand the class-membership of previous customers is well known and can be used in a classification model by learning both the information about the input and output values of each „old“ customer. (Supervised Learning).

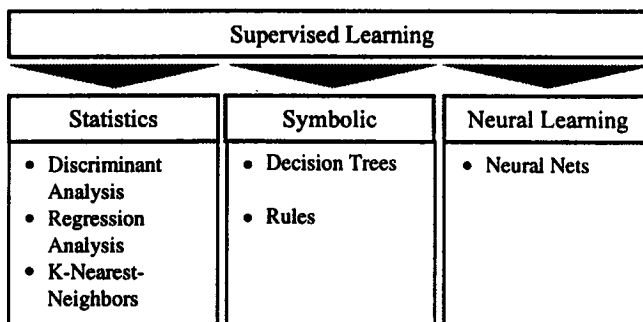


Figure 1: Alternative methods to handle credit scoring as classification task

These classification models are often built with the help of conventional, multivariate statistics, but as Fig. 1 shows, there are also methods and algorithms developed in the

field of artificial intelligence, e.g. machine learning and neural nets that offer appropriate solutions for the problem as well. [Michie et al. 1994, Graf and Nakhaeizadeh 1994]. The question which supervised learning method can be used in a particular case depends on the characteristics of the available data. On the other hand not all data sets can be processed by all methods, directly. Mostly, some additional work has to be done in advance before the learning process begins (pre-processing).

Statistical methods as discriminant analysis and k-nearest-neighbors, for example, require numerical input attributes. This is the same for neural nets.

The data we consider in this paper come from a European service provider in cellular phone business. The data sets contain that information each new customer is asked for, when he applies for a phone card that enables him to use his phone. In principal, the card inserted in the phone can be characterized as a credit card and has the same functionality: The cost of each phone call is charged to the customer's account and after a period of time (normally one month) he gets a detailed phone bill he has to pay.

In our case the information the service provider gets to decide on the credibility of a new customer is qualitative. It means that most of the supervised learning methods mentioned above cannot be used directly for classification with exception of symbolic learning methods that are able to deal with both qualitative and numerical inputs. For all other cases, attributes with qualitative values first have to be transformed in a numerical continuous form.

In the next section we consider different approaches of data transformation and apply them to our data of cellular phone customers.

Data Transformation

Generally, data transformation methods applicable to our problem can be divided in two groups: *Coding* [Lautsch 1993] and *scaling* approaches [Nishisato, 1994]. The idea of coding is to transform the qualitative information in a numerical code, in general by adding new attributes to the original data set. Scaling approaches, on the other hand, reproduce the original information on a numerical scale of a certain range.

In the following discussion we concentrate on one coding- and two slightly different scaling approaches and apply them to our raw data set.

Binary Coding

A very simple way to transform the information of a qualitative categorical attribute in a numerical is to code the information in a certain number of binary (0,1) variables.

Assume we got a categorical attribute j with K different values. Then for each possible value x_{jk} a new binary variable J_k is generated. The new variable J_k is assigned with a '1', if the value of the original categorical attribute j in the data set i is x_{jk} , otherwise the value of J_k is '0'. This

step has to be repeated for each data set i , for each categorical attribute j . At the end of that procedure the original categorical attributes are removed from the database.

This procedure can be written down in the following pseudo code:

```

FOR EACH data set i
  FOR EACH categorical attribute j
    FOR EACH possible value  $x_{jk}$ 
      CREATE new binary attribute  $J_k$ 
      IF  $x_{ij} = x_{jk}$ 
        THEN  $J_{ik} := 1$ ;
        ELSE  $J_{ik} := 0$ ;
      END; (*IF*)
    END; (*FOR*)
  ERASE categorical attribute j;
END; (*FOR*)
END; (*FOR*)

```

Example:

customer	attr. j (symbolic)	attr. j A (binary)	attr. j B (binary)	attr. j C (binary)	attr. j D (binary)
customer 1	A	1	0	0	0
customer 2	C	0	0	1	0
customer 3	D	0	0	0	1
customer 4	B	0	1	0	0
customer 5	D	0	0	0	1
customer 6	C	0	0	1	0
customer 7	A	1	0	0	0
...	...	...	...	...	...
customer N	B	0	1	0	0

In the example shown above 4 new binary attributes have been generated for the 4 values (A,B,C,D) of the categorical attribute, the information has been transferred in a binary code.

As we mentioned above, binary coding is a simple way to transform qualitative information in numerical shape. The main drawback of this method is that the number of attributes we have to process increases the more different values we got in the original data set. Using this approach, the number of attributes in our credit data rises from 16 categorical to 112 binary.

Univariate Scaling

Other possibilities to transform the given data sets in numerical continuous form are provided by several scaling approaches. These methods transform each categorical value in one numerical continuous value, so that the number of attributes in the data matrix does not change. The main advantage of such approaches is that the arrangement of the categories on the numerical scale gets an interpretable sense. This point becomes clearer at the end of this section.

In our particular case, the customers of cellular phone services are divided in two risk classes 'good' and 'bad'.

The class membership of 'old' customers is known and a very useful information for the scaling process. Nishisato [1994] analyses the dependencies between the class membership and each input attribute with the help of contingency tables and constructs the scale values for an attribute j with L categorical values in the following way (optimal scaling):

$$y_{jk} = \frac{p_{jk}^{good}}{p_{jk}^{good} + p_{jk}^{bad}} ; \quad k = 1, 2, \dots, L$$

for: y_{jk} scale value of category k in attribute j
 p_{jk}^{good} probability for being a *good* customer coming from category k of attribute j
 p_{jk}^{bad} probability for being a *bad* customer coming from category k of attribute j

In real world applications, generally, the probabilities are unknown and have to be substituted by frequencies we observe in the training data set.

In other words: We consider for each category k of an attribute j the quotient of good customers to all customers of that category as scale value for the category k .

Example: 100 customers,
attribute j with categories k_1, k_2, k_3

Contingency table:

j	k_1	k_2	k_3	Σ
good	20	15	10	45
bad	10	20	25	55
Σ	30	35	35	100

Scale values:

$$y_{jk_1} = \frac{0.20}{0.20 + 0.10} = 0.667$$

$$y_{jk_2} = \frac{0.15}{0.15 + 0.20} = 0.429$$

$$y_{jk_3} = \frac{0.10}{0.10 + 0.25} = 0.286$$

Note that the scale values get here an interpretable sense: The higher the scale value, the higher is the share of good customers coming from a specific category k .

All scale values are situated in the interval $[0;1]$, they become '0', if all customers coming from a category k of attribute j belong to the bad risk class ($p_{jk}^{good} = 0$), and '1', if all customers coming from a category k of attribute j

belong to the good risk class ($p_{jk}^{bad} = 0$).

Another quite similar scaling method has been developed by Fahrmeir et al. [1984] as shown below:

$$y_{jk} = \begin{cases} \frac{p_{jk}^{good}}{p_{jk}^{bad}} - 1 & \text{if } p_{jk}^{good} \geq p_{jk}^{bad} \\ 1 - \frac{p_{jk}^{bad}}{p_{jk}^{good}} & \text{otherwise} \end{cases}$$

Applying that to our example we get the scale values:

$$y_{jk_1} = \frac{0.20}{0.10} - 1 = 1.000$$

$$y_{jk_2} = 1 - \frac{0.20}{0.15} = -0.333$$

$$y_{jk_3} = 1 - \frac{0.25}{0.10} = -1.500$$

Here the range of scale values is not restricted to an certain interval (e.g. $[0;1]$). This approach has, however, an obvious and essential drawback: If any probability is estimated as '0', the scale value for the category is undefined.

Both scaling methods applied to our data return a new data matrix containing 16 numerical input attributes and one (non-manipulated) categorical output variable (risk class). In contrast to the method of binary coding the number of attributes remains constant.

Experimental Results

For the evaluation of the data transformation methods discussed in the previous section we applied them to our real world credit data and used the new numerical data sets to learn different classification models.

The raw data of 31.049 customers has been divided in a training data set of 16.376 examples for the learning process and in a test data set of 14.673 examples, we used for the performance evaluation on unseen cases. In both data sets the population is complete (no customer has been rejected), the portion of bad customers is at 9%.

Following classification methods have been used for our experiments:

- C4.5 has been developed by Quinlan [1993] and belongs to the group of symbolic machine learning algorithms. C4.5 generates either a decision tree or a set of classification rules directly from a raw data set of both categorical and numerical attributes. Here we needed no data transformation at all.
- Linear discriminant analysis (LDA), quadratic discriminant analysis (QDA) and k-nearest-neighbors (KNN) belong to the group of classification methods coming from conventional multivariate statistics. We

have used the versions implemented in SAS®.

- **DIPOL** is a hybrid linear classifier developed by Schulmeister et al. [1997] combining the idea of pairwise linear regression with an optimization by neural nets.
- As neural net (NN) we trained a full-linked backpropagation net of 112 input, 20 hidden and 2 output neurons in the case of binary coded data, in the case of the scaled data we chose 16 input, 4 hidden and 2 output neurons.

In a first step we determined the minimum error rate each classifier reached on the test data for the different data transformation approaches:

Supervised Learning Method	Error rate			
	Raw data	Binary data	Scaled data	
			Nishisato	Fahrmeir
C4.5	4,59%	4,93%	4,99%	5,01%
LDA	-	5,70%	5,70%	5,70%
QDA	-	5,95%	5,89%	5,85%
KNN	-	6,20%	5,17%	5,46%
DIPOL	-	5,09%	4,68%	4,79%
NN	-	4,71%	4,65%	4,68%

Comparing the error rates we observe that in most cases the optimal scaling approach [Nishisato, 1994] dominates the other data transformation methods, on binary data we rarely reached the results we got on scaled data. All in all the symbolic learning algorithm C4.5 processed on raw data showed the lowest error rate.

In a second step we observed the time each learning process spent to build a classification model. This criterion gets particularly important, if the classifier has to be updated from time to time caused by dynamic changes in real world data.

Supervised Learning Method	Time spent for learning process			
	Raw data	Binary data	Scaled data	
			Nishisato	Fahrmeir
C4.5	20 s	60 s	40 s	40 s
LDA	-	2.0 min	1.0 min	1.0 min
QDA	-	2.5 min	1.5 min	1.5 min
KNN	-	5 h	18.0 min	18.0 min
DIPOL	-	2.0 min	40 s	40 s
NN	-	5 h	1 h	1 h

Here, the essential drawback of the binary coding method is very obvious: Strongly increasing number of attributes slows down the learning process in a quite unacceptable way. In circumstances where the learning phase should be often repeated (for example, to adjust the learning parameters) the scaling approaches are more suitable.

Conclusion

In this paper we considered different approaches of data transformation and evaluated them on real world credit data by using several supervised learning methods.

In general, both the binary coding and the scaling approach offer appropriate solutions for transforming categorical in numerical data. Concerning the error rate the scaled data sets always reached better results than binary coded data sets. The advantage of scaling becomes, however more obvious, when we consider the learning time aspects. The increasing number of input attributes in binary data caused a heavy slow down of the learning processes.

All in all the symbolic learning algorithm C4.5 returned the best results in error rate and time spent for the learning process. It means that in our study the efforts put to coding and scaling of the categorical attributes did not contribute to improve the results. We can conclude that the best way to handle the categorical attributes in credit scoring domains is to use learning algorithms that are able to deal with such kind of attributes directly without data transformation.

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